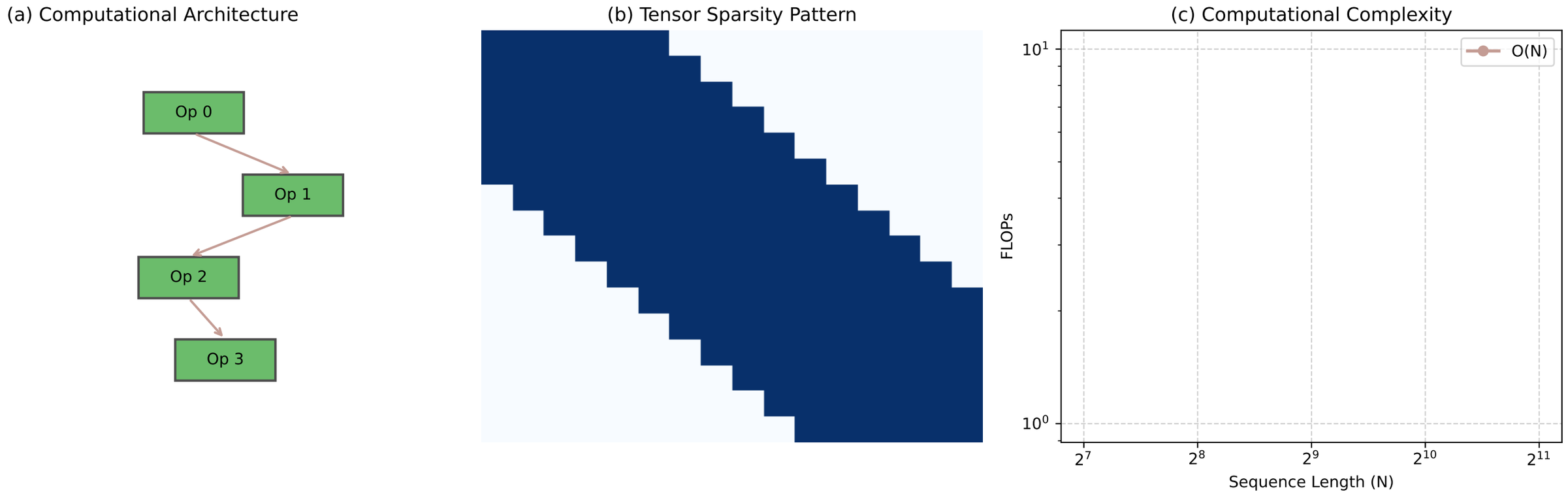Module Architecture: preprocessing